Interpreting Agent Behavior (IAB) Workshop @ NeurIPS 2026 · GLEE Competition Track

A Cost-and-Behavior Case Study of Two GLEE Agents with Design-Time and Runtime Behavioral Interpretation

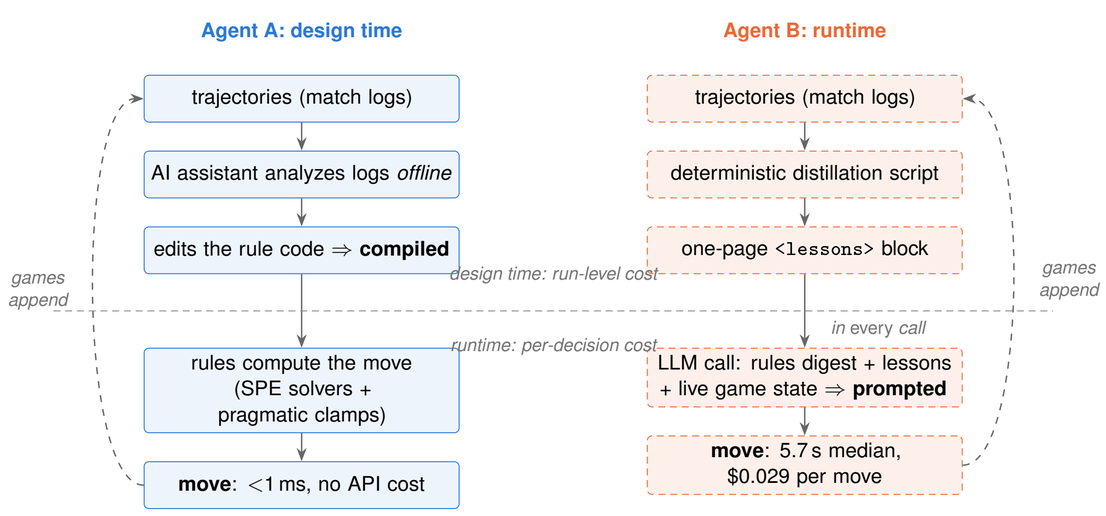
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code & aggregated data

Two of our GLEE entries learn from the same kind of evidence, their own match logs, but use it at opposite ends of the stack. **Agent A** compiles the lessons into rules at *design time*. **Agent B** prompts an LLM with them at *runtime*.

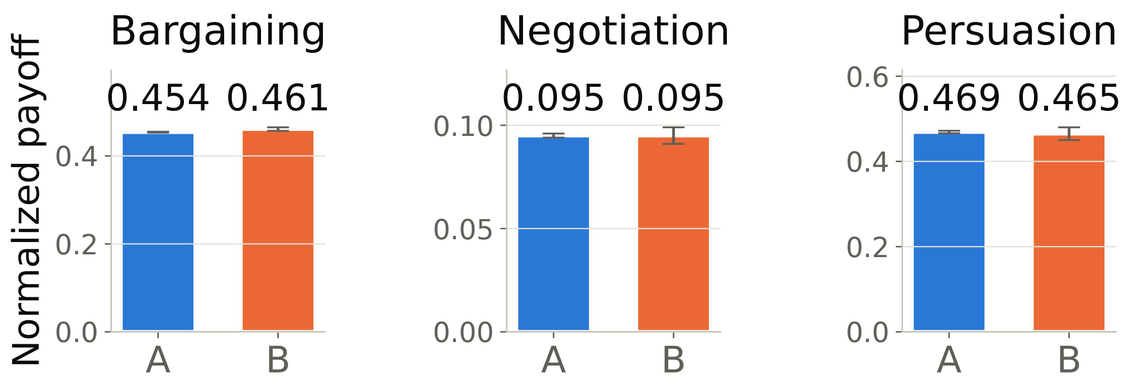
What did each deployment return, what did it cost, and how did it behave?

Two agents, one interpretation loop



- **A** (`Rubinstein`): subgame-perfect bargaining solvers, reservation-value negotiation, Bayesian-persuasion lying set 25% below the commitment-optimal rate. An AI assistant made ∼15 rule edits over ten days.
- **B** (`Rubinstein-RT`): every move is one LLM call with a `<lessons>` block distilled from B's own logs. Invalid or timed-out calls fall back to A's rules (0.9% of moves).

Returns: similar point estimates



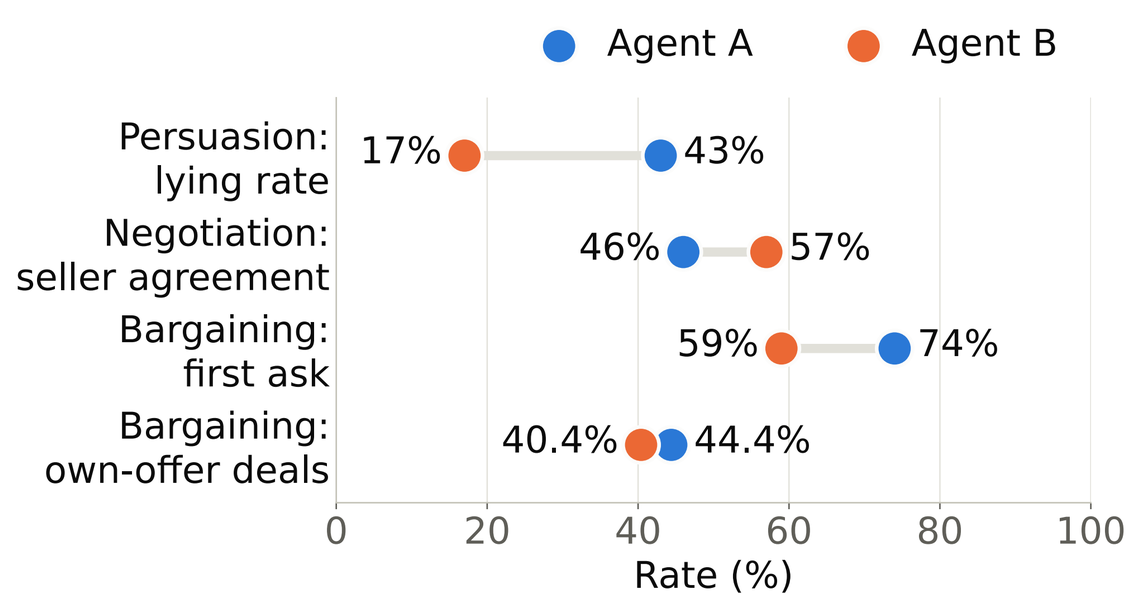
Mean normalized payoff ± SE. B−A differences: **+0.008** bargaining, **0.000** negotiation, **-0.005** persuasion. Games: A 120,968, B 4,107 (29× fewer). SEs treat games as independent, so they describe spread only. We make no significance or equivalence claim.

Costs: run-level vs. per-decision

$151	**none**	**$1,158**
A: design time	A: runtime model API	B: runtime LLM layer
(AI build + monitoring)	(<1 ms per move)	($0.28/game, 63,238 calls)

B's build cost $86; its successful calls took 80 inference hours (median 5.7 s, p90 13.1 s per move). At A's volume, B would cost ≈$34k. These are model-API figures at August-2026 prices; human time and hosting are excluded.

Agent Behavior Analysis: different behaviors



Each row has its own denominator: low-quality persuasion rounds, seller games, games with an observed first ask, and bargaining agreements.

The LLM lied far less than its brief allowed. A lied on 43% of low-quality persuasion rounds, as designed. B was briefed with a ceiling but no target rate. It lied on 17%, about 40% of A's rate, and withheld recommendations on 36% of high-quality rounds.

Two points on the volume–margin frontier. As a negotiation seller, A agreed in 46% of games at a median 1.19× its valuation. B agreed in 57% at 1.15×. B also settled bargains earlier (median round 2 vs. 3).

Terminal outcomes change the acceptance picture. Pre-action histories showed zero accepted own offers, but only because no action is due after an acceptance. Joining server outcomes shows that own offers closed 44.4% (A) and 40.4% (B) of agreements.

Takeaways

- Similar payoffs can hide different agreement, margin, and persuasion strategies. Trajectories reveal them, but only if terminal outcomes are included.
- Compiled rules give near-zero runtime cost and code-level predictability, but adapting them takes explicit revision.
- Runtime LLM control allows more discretion, costs real inference money, and does not guarantee that soft guidance is followed.

Caveat. This is an observational case study of two unmatched, evolving deployments (different volumes, periods, pools, and base strategies). It generates hypotheses and identifies no causal effects.

GLEE: Shapira et al., 2024 · Bayesian persuasion: Kamenica & Gentzkow, 2011 · Rubinstein, 1982

`https://github.com/starchou6/glee-iab-2026` (commit cffe01f)